

Smart Meter

Electric Metering Specifications

Rated Current	5(80) A
Rated Voltage	220V / 240V
Frequency	50 / 60 Hz
Accuracy Class	Class 1 / Class D
Impulse Constant	1600 imp/kWh
Operating Temp	-40°C to +125°C
Display	2.5" Touch Screen
Max Load	100A Mains / 60A Geyser

Construction

Materials & Build

Material	ABS+ (20 to 4B0)
Dimensions	160 x 160 x 45 mm
Contacts	Silver Tin Oxide (AgSnO2)
Switching Power	25,000 VA / 440 VAC
Mechanical Life	1,000,000,000 Cycles

Standards

International Compliance

- IEC 62053-21 / IEC 62053-23
- EN50470-1 / EN50470-3
- OIML R46 & ANSI C12.20
- IEC 62053-24 (Reactive)
- STS Token Specification (IEC 62055-41)

Key Features

Advanced Metering Capabilities

Autocalibration Automatic calibration based on direct measurement of the full signal path. No reference meter required.
3x High-Perf ADCs 86 dB SNR with 16-bit resolution for precise energy measurement across all phases.
Advanced Metrology WATT, VAR, VA, Wh, VARh, VAh measurement. Current and voltage RMS. Full power quality analysis.
STS Token Security Validates, decrypts, authenticates tokens using international STS standards. Tamper-proof key management.
Remote Management Token transfer via SMS, BLE, Web platform, or direct touch input. Full remote load control.
Geyser Control Manual on/off, daily scheduling with day selection, countdown timer. BLE app or touch screen control.

Energy Monitoring

Real-Time Tracking

- 24-hour usage chart (hourly bars in kWh)
- Dynamic scaling & daily reset at midnight
- Live current, voltage, power & temperature
- Power factor & reactive power monitoring
- Consumption data persists across restarts
- 30-second auto-refresh for real-time view

Integrated Router

Built-In Smart Wi-Fi System

Router Hardware Specifications	
RAM	DDR2 64 MB
Flash	SPI 16 MB (max 32 MB)
Wi-Fi Standard	IEEE 802.11 b/g/n
Wireless Speed	150 Mbps
Frequency	2.4 GHz / 5 GHz
Antennas	2x FPC Wi-Fi + 2x FPC 4G
WAN Port	1x 10/100M (Auto MDI/MDIX)
LAN Port	1x 10/100M (Auto MDI/MDIX)
Power	POE Power Supply
Cellular Module	Quectel EC200A (4G LTE)

Network Modes

Flexible Deployment

Client Mode	Connect to existing wireless network
Repeater	Extend coverage range wirelessly
Bridge Mode	Multiple SSIDs & encryption modes
Router Mode	Full routing with DHCP server
AP Client Router	Share via WAN/VPN passthrough
Cellular (4G)	Mobile network SIM card access

Cellular (4G/LTE)

Always-On Connectivity

- Module: Quectel EC200A (4G LTE Cat-1)
- Band: FDD LTE Band 3
- Network: 4G with auto-failover
- SIM: Standard SIM slot (Normal status)
- Signal Metrics: RSRP, RSRQ, RSSI, SINR
- Remote management via cellular backhaul

Advanced Networking

Enterprise-Grade Features

- Port Forwarding & DMZ
- DDNS & VPN Client (OpenVPN)
- QoS Traffic Management
- DTU (Data Terminal Unit)
- Firewall & Web Access Control
- Print Server & Network Storage
- Client Management (MAC/IP/IRSSI)
- Blacklist & Access Control

Software Platform

GRIDx Utility Management Portal

Web Dashboard Full utility management portal at gridx-meters.com. Real-time monitoring of entire meter fleet.
Mobile App Android BLE app for local meter control. Geyser scheduling, token entry, power monitoring.
Remote Monitoring Real-time network monitoring with alerts & notifications. Complete grid topology visualization.
Analytics & Reports Meter accuracy tracking, load analysis, consumption trends, billing reports for utilities.

Emergency Services

One-Touch Safety Buttons

Police Two-press emergency alert to authorities	
Ambulance Medical emergency request & dispatch	
Fire Brigade Fire emergency alert with confirmation	

Token Processing

Secure Credit Management

- STS standard token authentication
- 20-digit encrypted token validation
- Multi-channel: SMS, BLE, Web, Keypad
- Tamper detection & rejection logging
- Auto key-change & proprietary commands
- Full transaction history with timestamps

SMS Control

Remote Access Management

- Up to 5 authorized phone numbers
- SMS-based remote commands
- Status updates via SMS broadcast
- International format support
- Add/remove via touch screen or web

